LAPORAN PRAKTIKUM
ADMINISTRASI DAN MANAJEMEN JARINGAN

UNIT 2
SAMBA SERVER



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BAB 1 PENDAHULUAN

A. Latar belakang

Samba is a free and open-source software that provides an implementation of the SMB/CIFS protocol for non-Windows operating systems such as Linux, Unix, and macOS [1]. A Samba server allows these operating systems to function as a file and print server in a Windows network environment, enabling Windows users to share and access files and printers from computers running non-Windows operating systems, as illustrated in Figure 2.1. Samba provides features such as authentication, authorization, encryption, and access rights management to ensure the security and privacy of shared data. It can also be integrated with directory services such as Active Directory to simplify user management in large network environments [2].

Samba operates using Server Message Block (SMB) protocol — a network protocol used to share files, printers, and other network resources between Windows or Unix computers [3]. SMB was originally developed by Microsoft as part of LAN Manager in the 1980s and has since become the de facto standard for file and printer sharing in Windows networks. SMB is a client-server protocol: the client sends requests to the server, and the server responds by delivering data or performing the requested operation, as illustrated in Figure 3.1. Data exchange is secured using encryption, authentication, and authorization.

Samba consists of two programs running in the background:

- SMBD — the file server; spawns a new process for each active client.
- NMBD — converts NetBIOS computer names to IP addresses and monitors shared resources on the network.

SMBD behavior is controlled through /etc/samba/smb.conf [4]. Samba can be configured as a file server, print server, or domain controller.

B. Tujuan

1. Students are able to install Samba Server.
2. Students are able to modify Samba configuration to manage user access rights and shared resources.
3. Students are able to practice file sharing between Ubuntu and Windows using Samba.

BAB 2 PEMBAHASAN

A. Langkah praktikum

1. Samba Server Installation

a. Update the package list

```
vboxuser@Ubuntu-gw:~$ sudo apt update
Hit:1 http://id.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://id.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://id.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
101 packages can be upgraded. Run 'apt list --upgradable' to see them.
vboxuser@Ubuntu-gw:~$
```

b. Install Samba

```
vboxuser@Ubuntu-gw:~$ sudo apt install samba
samba is already the newest version (2:4.23.6+dfsg-1ubuntu2.2).
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-14          linux-image-unsigned-7.0.0-14-generic    linux-modules-7.0.0-14-generic  linux-tools-7.0.0-14-generic
  linux-headers-7.0.0-14-generic  linux-main-modules-zfs-7.0.0-14-generic  linux-tools-7.0.0-14
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 101
vboxuser@Ubuntu-gw:~$
```

c. Verify the installation

```
vboxuser@Ubuntu-gw:~$ systemctl status smbd
● smbd.service - Samba SMB Daemon
     Loaded: loaded (/usr/lib/systemd/system/smbd.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-09-01 05:38:51 UTC; 1min 15s ago
 Invocation: fd742c5ac3054d068e7ec59744bf3761
       Docs: man:smbd(8)
             man:samba(7)
             man:smb.conf(5)
   Main PID: 2089 (smbd)
     Status: "smbd: ready to serve connections..."
      Tasks: 3 (limit: 5992)
     Memory: 28.5M (peak: 28.8M)
        CPU: 77ms
     CGroup: /system.slice/smbd.service
             ├─2089 /usr/sbin/smbd --foreground --no-process-group
             ├─2093 "smbd: notifyd" .
             └─2094 "smbd: cleanupd "
vboxuser@Ubuntu-gw:~$
```

2. Firewall Setup

a. Check firewall status & Allow Samba traffic through the firewall

```
vboxuser@Ubuntu-gw:~$ sudo ufw status
[sudo: authenticate] Password:
Status: inactive
vboxuser@Ubuntu-gw:~$ sudo ufw enable
Firewall is active and enabled on system startup
vboxuser@Ubuntu-gw:~$ sudo ufw allow samba
Rule added
Rule added (v6)
vboxuser@Ubuntu-gw:~$
```

B. Exercise

1. Setting Up File Sharing

a. Create the shared directories

```
vboxuser@Ubuntu-gw:~/prak-amj/week2$ mkdir share1
vboxuser@Ubuntu-gw:~/prak-amj/week2$ mkdir share2
```

```
vboxuser@Ubuntu-gw:/home$ sudo mv /home/vboxuser/prak-amj/week2/share1 /home/
mv: cannot stat '/home/vboxuser/prak-amj/week2/share1': No such file or directory
vboxuser@Ubuntu-gw:/home$ sudo mv /home/vboxuser/prak-amj/week2/share2 /home/
mv: cannot stat '/home/vboxuser/prak-amj/week2/share2': No such file or directory
vboxuser@Ubuntu-gw:/home$
```

b. Create an empty test file inside each directory:

```
vboxuser@Ubuntu-gw:~/prak-amj/week2$ touch share1/testfile1.txt
vboxuser@Ubuntu-gw:~/prak-amj/week2$ touch share2/testfile2.txt
```

c. Set directory permissions

```
vboxuser@Ubuntu-gw:/home$ chmod 777 /home/share1
vboxuser@Ubuntu-gw:/home$ chmod 777 /home/share2
```

d. Create Samba users

```
vboxuser@Ubuntu-gw:~$ sudo useradd user1
vboxuser@Ubuntu-gw:~$
vboxuser@Ubuntu-gw:~$ sudo useradd user2
```

e. Set Samba passwords

```
vboxuser@Ubuntu-gw:~$ sudo smbpasswd -a user1
New SMB password:
Retype new SMB password:
Added user user1.
vboxuser@Ubuntu-gw:~$ sudo smbpasswd -a user2
New SMB password:
Retype new SMB password:
Added user user2.
vboxuser@Ubuntu-gw:~$
```

f. Configure access rights in smb.conf

```
[share1]
path = /home/share1
valid users = user1, user2
read list = user2
write list = user1
browseable = yes

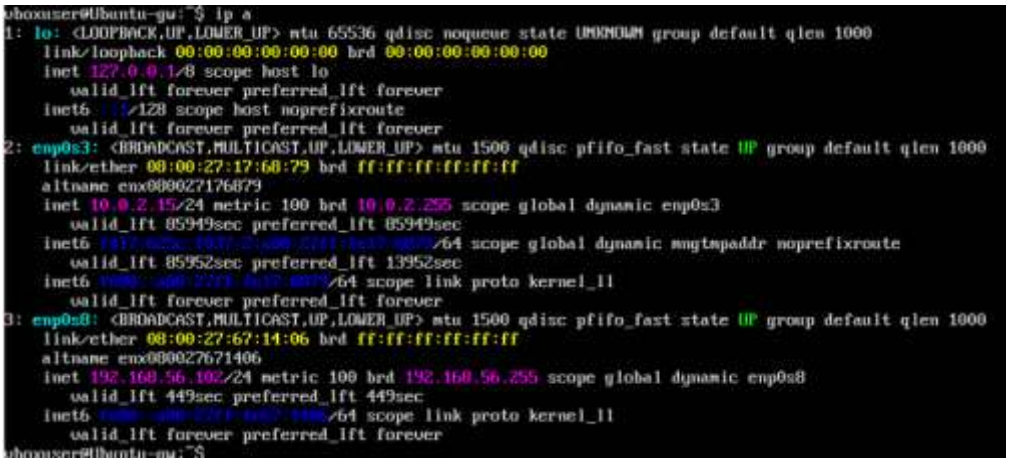
[share2]
path = /home/share2
valid users = user1, user2
read list = user1
write list = user2
browseable = yes
```

g. Restart Samba

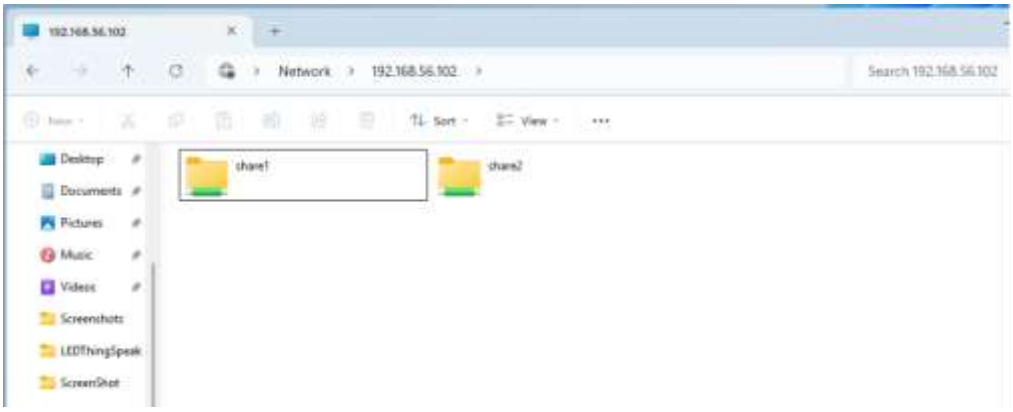
```
vboxuser@Ubuntu-gw:~$ systemctl restart smbd
==== AUTHENTICATING FOR org.freedesktop.systemd1.manage-units ====
Authentication is required to restart 'smbd.service'.
Authenticating as: vboxuser
Password:
==== AUTHENTICATION COMPLETE ====
vboxuser@Ubuntu-gw:~$
```

2. Testing File Sharing from Windows

a. Find the Ubuntu Server IP address



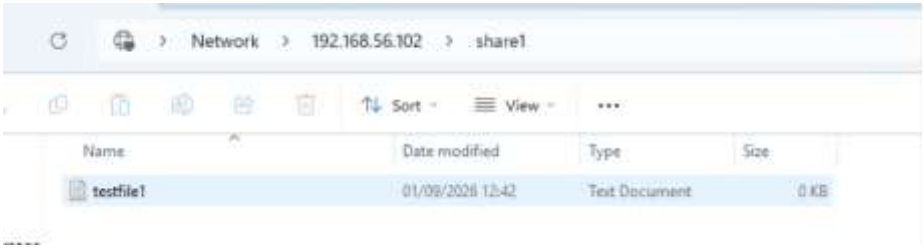
b. Open the shares in File Explorer



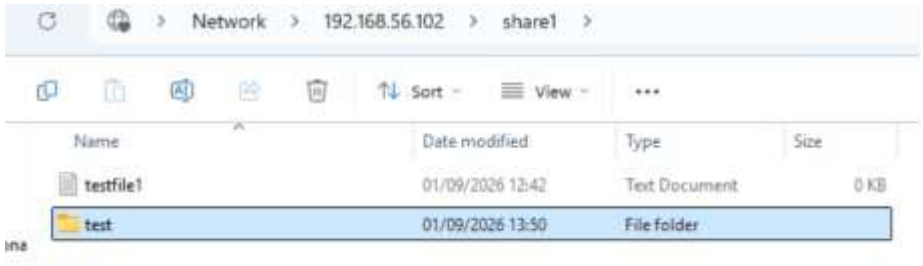
c. Test access as user1

Log in as user1

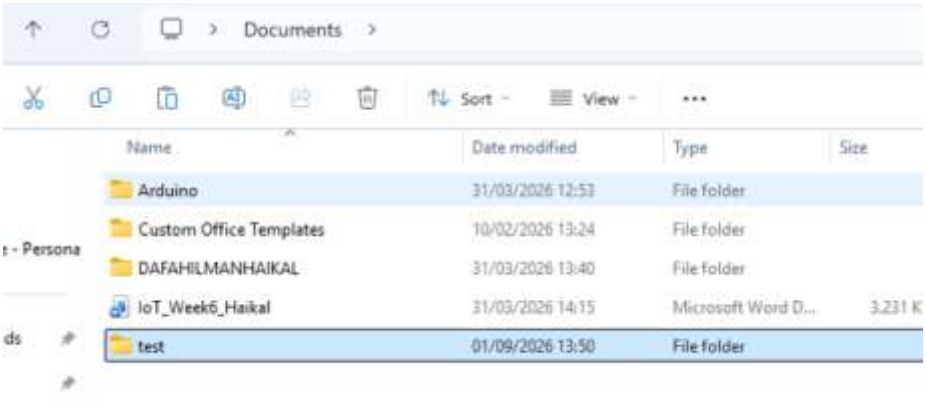
- Can you view the contents of share1?



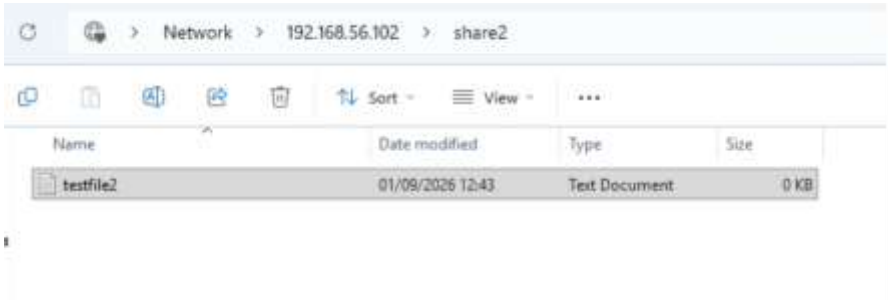
- Can you create a new file inside share1?



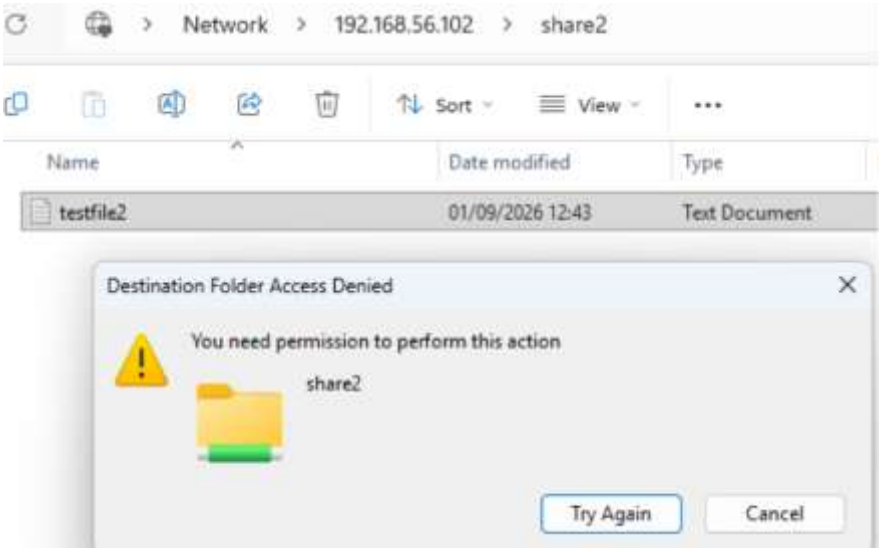
- Can you copy a file from share1 to local Windows storage?



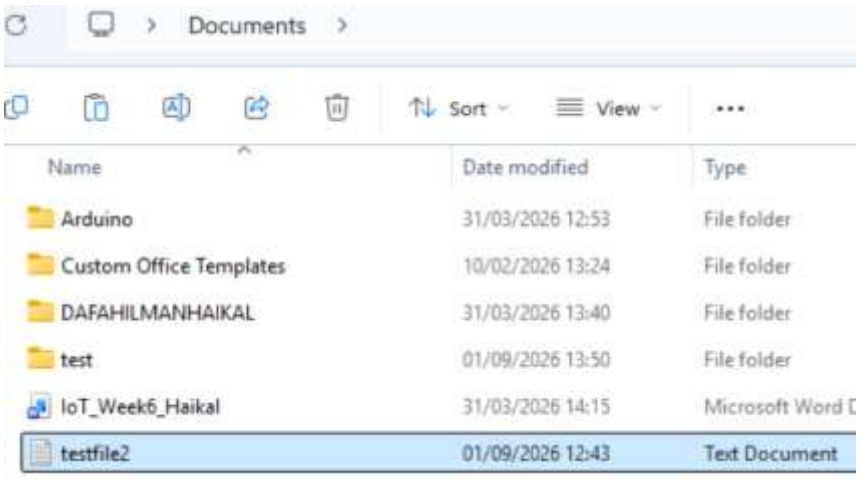
- Can you view the contents of share2?



- Can you create a new file inside share2?



- Can you copy a file from share2 to local Windows storage?



d. Disconnect and switch to user2

```
C:\Users\TAJ>net use
New connections will be remembered.

Status       Local     Remote                    Network

-------------------------------------------------------------------------------
OK                     \\192.168.56.102\share1   Microsoft Windows Network
OK                     \\192.168.56.102\share2   Microsoft Windows Network
OK                     \\192.168.56.102\IPC$     Microsoft Windows Network
The command completed successfully.


C:\Users\TAJ>net use \\192.168.56.102\ /delete
\\192.168.56.102\ was deleted successfully.


C:\Users\TAJ>net use
New connections will be remembered.

Status       Local     Remote                    Network

-------------------------------------------------------------------------------
OK                     \\192.168.56.102\share1   Microsoft Windows Network
OK                     \\192.168.56.102\share2   Microsoft Windows Network
The command completed successfully.
```

```
C:\Users\TAJ>net use
New connections will be remembered.

Status       Local     Remote                    Network

-------------------------------------------------------------------------------
OK                     \\192.168.56.102\share1   Microsoft Windows Network
OK                     \\192.168.56.102\share2   Microsoft Windows Network
The command completed successfully.

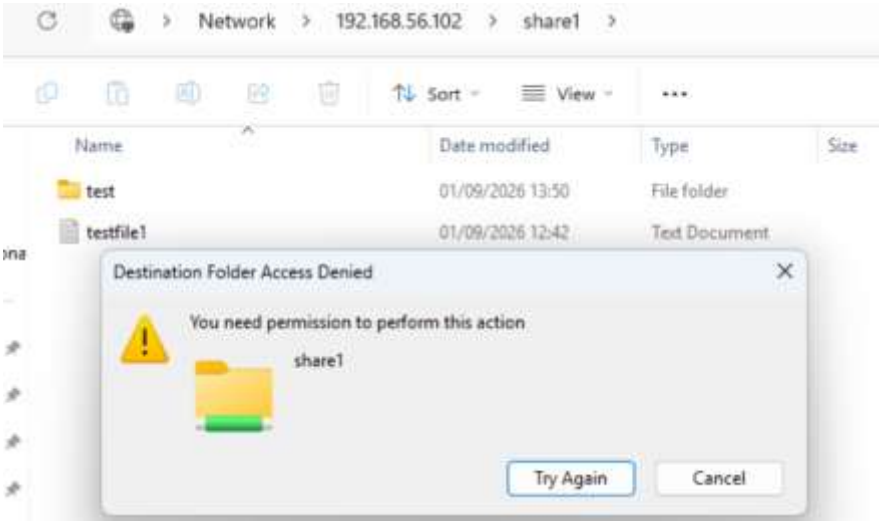

C:\Users\TAJ>net use \\192.168.56.102\share1 /delete
\\192.168.56.102\share1 was deleted successfully.
```

```
C:\Users\TAJ>net use \\192.168.56.102\share2 /delete
\\192.168.56.102\share2 was deleted successfully.
```

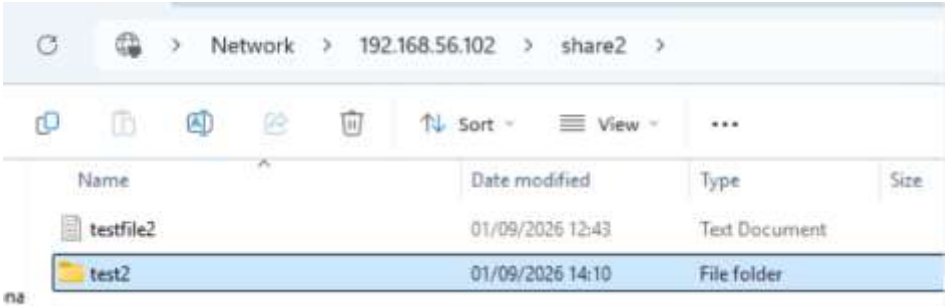
e. Test access as user2

```
C:\Users\TAJ>net use \\192.168.56.102\share1 /user:user2
Enter the password for 'user2' to connect to '192.168.56.102':
The command completed successfully.
```

- Create a new file inside share1



- Create a new file inside share2



C. Pembahasan

Pada praktikum ini, praktikan perlu melakukan instalasi dan konfigurasi Samba Server pada Ubuntu Server. Tahap pertama yang dilakukan yaitu memperbarui daftar paket dengan menggunakan perintah ‘sudo apt update’, kemudian melakukan instalasi Samba pada sistem. Setelah proses instalasi selesai, dilakukan verifikasi untuk memastikan Samba Server telah berhasil terpasang. Selanjutnya, dilakukan konfigurasi firewall dengan memeriksa status firewall dan mengizinkan trafik Samba agar layanan dapat digunakan untuk melakukan komunikasi dan berbagi file melalui jaringan.

Setelah Samba Server berhasil diinstal, praktikan kemudian melakukan konfigurasi file sharing. Beberapa direktori yang akan digunakan sebagai file sharing dibuat terlebih dahulu, kemudian masing-masing direktori diberikan file kosong (testfile) sebagai file pengujian. Setelah itu, permission terhadap direktori diatur untuk bisa mengeksekusi semuanya (chmod 777). Praktikan juga membuat beberapa user dan memberikan password Samba pada masing-masing user. Pengaturan hak akses dan direktori yang akan dibagikan kemudian dilakukan melalui file konfigurasi /etc/samba/smb.conf. Setelah seluruh konfigurasi selesai, layanan Samba direstart agar perubahan konfigurasi dapat berjalan.

Tahap berikutnya yaitu melakukan pengujian file sharing dari komputer Windows. Praktikan terlebih dahulu mengetahui alamat IP Ubuntu Server, kemudian mengakses direktori yang telah dibagikan dengan mengakses IP Ubuntu melalui File Explorer pada Windows. Pengujian dilakukan menggunakan user1 terlebih dahulu untuk mengetahui apakah pengguna tersebut dapat melihat isi share1 dan share2, membuat file baru pada direktori yang memiliki hak akses, serta menyalin file dari server ke penyimpanan lokal Windows. Setelah pengujian menggunakan user1 selesai, koneksi diputus dan diganti menggunakan user2. Pengujian kembali dilakukan untuk melihat perbedaan hak akses yang dimiliki user2, termasuk mencoba membuat file pada share1 dan share2. Hasilnya, ketika log in menggunakan user1, praktikan tidak bisa menambahkan file ke folder share1. Sedangkan ketika praktikan log in menggunakan user2, penambahan filie pada folder share2 tidak bisa dilakukan. Melalui pengujian tersebut, praktikan dapat mengetahui bahwa konfigurasi Samba tidak hanya digunakan untuk berbagi file antara Ubuntu dan Windows, tetapi juga dapat digunakan untuk mengatur akses setiap pengguna terhadap sumber daya yang dibagikan.

BAB 3 PENUTUP

A. Kesimpulan

Berdasarkan praktikum yang telah dilakukan, dapat disimpulkan bahwa Samba Server merupakan perangkat lunak berbasis open-source yang dapat digunakan untuk menyediakan layanan file sharing antara sistem operasi Linux dan Windows. Pada praktikum ini, Samba Server berhasil diinstal pada Ubuntu Server dengan melakukan pembaruan daftar paket, instalasi Samba, serta verifikasi instalasi. Selain itu, konfigurasi firewall juga dilakukan dengan mengizinkan trafik Samba sehingga layanan dapat digunakan untuk melakukan komunikasi dan berbagi file melalui jaringan.

Praktikum ini juga memberikan pemahaman mengenai konfigurasi hak akses pengguna dan sumber daya pada Samba Server. Praktikan membuat direktori share, file pengujian, serta beberapa pengguna Samba yang kemudian diberikan password dan hak akses melalui konfigurasi /etc/samba/smb.conf. Pengujian dilakukan melalui Windows menggunakan user1 dan user2 untuk mengakses share1 dan share2, membuat file, serta menyalin file ke penyimpanan lokal. Hasil pengujian menunjukkan adanya perbedaan hak akses antara kedua pengguna, sehingga dapat diketahui bahwa Samba dapat digunakan tidak hanya untuk melakukan file sharing, tetapi juga untuk mengatur akses pengguna terhadap sumber daya yang dibagikan.

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